

Manoj Avula
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PROFESSIONAL SUMMARY:

- Senior Data Engineer with 6+ years of experience in enterprise data engineering, large-scale ETL development, and cloud-based analytics solutions.
- Strong expertise in Azure Databricks, Azure Data Factory, Unity Catalog, Python, PySpark, and SQL for building scalable, secure, and high-performance data pipelines.
- Proven experience designing, deploying, and supporting production-grade data pipelines processing millions of records under strict SLA requirements.
- Extensive hands-on experience modernizing legacy ETL workflows and optimizing Spark jobs for improved performance and cost efficiency.
- Deep understanding of data warehousing concepts, dimensional modeling, incremental data loads, and scalable data architecture design.
- Expertise in implementing enterprise-grade data governance frameworks, metadata management solutions, and RBAC-based access controls.
- Strong background in data validation, reconciliation, audit reporting, and regulatory compliance across enterprise data platforms.
- Hands-on experience with ETL testing strategies, automation frameworks, and CI/CD-driven data pipeline deployments.
- Proven ability to design and implement secure data ingestion mechanisms, including API-based integrations and SFTP-based data transfers.
- Skilled in developing API-driven data platforms using FastAPI, Flask, and REST-based architectures.
- Extensive experience working with structured, semi-structured, and large-scale datasets across cloud and on-prem environments.
- Strong SQL expertise in complex joins, subqueries, aggregations, window functions, and query performance tuning.
- Experienced in production support, incident management, root-cause analysis, and resolution of data pipeline failures.
- Proven experience designing fault-tolerant and highly available data pipelines with retry logic, monitoring, and alerting mechanisms.
- Strong expertise in performance tuning and cost optimization across cloud data platforms while maintaining scalability and reliability.
- Hands-on experience working in Agile and Scrum environments, collaborating through sprint planning, reviews, and retrospectives.
- Demonstrated ability to own end-to-end data solutions from requirements gathering and design to deployment and production support.
- Strong documentation and knowledge-sharing skills, including technical design documents, runbooks, and operational guides.
- Ability to translate complex business requirements into scalable, efficient, and maintainable technical solutions.
- Effective communicator and mentor, collaborating across engineering, analytics, DevOps, and business teams.

TECHNICAL SKILLS:

Languages	Python, PySpark, SQL
Data Platforms	Azure Databricks, Apache Airflow, Azure Data Factory, Spark, Kafka
Governance & Security	Unity Catalog, RBAC, Access Control, Data Lineage, Metadata Management
Databases	MySQL, PostgreSQL, MongoDB, Snowflake
Storage	Azure Data Lake Storage Gen2, Amazon S3
APIs	FastAPI, Flask, REST APIs
Cloud & DevOps	Microsoft Azure, Docker, CI/CD Pipelines, Azure DevOps, GitHub Actions
Testing & Quality	ETL Testing, Data Validation, Data Reconciliation, Audit Logging
BI & Tools	Tableau, Power BI, MS Excel, Git, Jira, Postman, Linux/Unix

PROFESSIONAL EXPERIENCE:

Johnson & Johnson - Raritan, NJ

01/2025 - Present

Role: Senior Data Engineer

Responsibilities:

- Designed and implemented enterprise-scale ETL pipelines using Azure Data Factory to ingest, transform, and load data into Azure Data Lake Storage Gen2 from multiple internal and external sources.
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- Led the migration and enterprise rollout of Unity Catalog, enabling centralized metadata management, fine-grained RBAC access controls, and secure data sharing across Databricks workspaces.
- Refactored and optimized legacy Databricks notebooks and Spark workloads, improving execution performance by up to 90% while reducing compute costs.
- Developed and optimized PySpark-based data transformations to improve overall data processing throughput and efficiency.
- Processed and validated 5M+ records daily, ensuring high data accuracy, consistency, and availability for downstream analytics use cases.
- Implemented secure SFTP-based ingestion pipelines for partner and vendor data in compliance with enterprise security and regulatory standards.
- Built comprehensive data quality checks, reconciliation logic, and audit controls to proactively identify and reduce data defects.
- Created and executed ETL test cases, validation reports, and production runbooks to support operational readiness and support teams.

- Validated data consistency between source and target systems using SQL-based reconciliation techniques and external tables.
- Supported automation-driven data extraction and enrichment workflows to improve ingestion efficiency and reduce manual effort.
- Conducted end-to-end testing of ETL workflows and data mappings to validate data integrity across systems.
- Performed functional, regression, and UAT testing for ETL workflows and data integrations prior to production releases.
- Maintained detailed technical documentation covering ETL logic, dependencies, edge cases, and operational procedures.
- Collaborated closely with data architects and business stakeholders to deliver scalable, compliant, and high-performing data solutions.
- Mentored junior engineers on ETL best practices, code reviews, and Azure cloud deployment standards.

Environment: Python, Azure Databricks, Unity Catalog, ADLS Gen2, PySpark, SQL, Python, Git, Azure DevOps, ETL Testing.

Trumble Inc - Farmington Hills, MI

01/2023 – 12/2024

Role: Data Engineer

Responsibilities:

- Designed and deployed end-to-end ETL workflows using Python and Apache Airflow to process and transform large-scale datasets.
- Tested, scheduled, and troubleshoot ETL jobs to ensure reliable, timely, and error-free pipeline execution.
- Designed and implemented incremental and batch processing strategies to efficiently handle increasing data volumes.
- Built and maintained RBAC-based data access controls to secure sensitive datasets and enforce compliance standards.
- Designed and optimized PostgreSQL schemas and complex SQL queries to improve performance and scalability.
- Developed reusable Python modules and standardized Airflow DAG templates to streamline ETL development.
- Implemented data validation and reconciliation checks to ensure accuracy between source systems and target databases.
- Collaborated with product and analytics teams to define data contracts, schemas, and consumption patterns.
- Enhanced pipeline reliability by implementing retry mechanisms, alerting, and robust failure-handling logic.
- Performed root cause analysis on data issues and implemented permanent fixes to reduce recurring incidents.
- Supported environment promotions from development to test and production while ensuring configuration consistency.
- Documented ETL workflows, API contracts, and operational runbooks to improve onboarding and support efficiency.

- Assisted in performance tuning of PostgreSQL queries using indexing strategies and query optimization techniques.
- Developed FastAPI-based REST services to expose curated and validated datasets for internal applications.
- Integrated automated data extraction workflows for structured and semi-structured data sources.
- Improved ETL performance by approximately 30% through code optimization and scheduling enhancements.
- Implemented multi-format data exports (CSV, JSON, TXT) with validation logic for downstream analytics.
- Monitored and troubleshoot production pipelines to ensure high availability and SLA compliance.
- Supported CI/CD-driven deployments for data pipelines and APIs to improve release stability.

Environment: Python, Airflow, FastAPI, PostgreSQL, Docker, Git.

CVS - Dallas, TX

01/2022-01/2023

Role: Data Engineer

Responsibilities:

- Coordinated with business and technical stakeholders to gather requirements and deliver data solutions supporting analytics and operations.
- Implemented and managed large-scale data storage solutions using Azure Data Lake Storage.
- Developed distributed data processing workflows using Apache Spark on Azure Databricks.
- Integrated Azure Data Factory with Azure Databricks, Azure Synapse Analytics, and Azure SQL Database.
- Developed complex ETL pipelines to automate data ingestion, transformation, and loading from diverse data sources.
- Utilized Spark, Scala, and Python to query, transform, and prepare data from large-scale data sources.
- Created Hive external tables and managed data storage in HDFS using ORC, Parquet, and Avro formats.
- Implemented Sqoop jobs to import data from RDBMS systems into HBase using incremental and full refresh strategies.
- Developed Python frameworks to convert Hive and SQL queries into Spark transformations using RDDs and Spark SQL.
- Designed data warehouse structures using Hive external tables for job control, monitoring, and data quality tracking.
- Developed Spark Scala frameworks to read data from Hive and write aggregated results to HBase.
- Managed and scheduled workflows by defining Hive, Spark, Sqoop, and Python actions using Oozie workflows and Oozie Coordinator.

Environment: Spark, Scala, Hadoop, Python, Pyspark, Azure, MapReduce, ETL, HDFS, Hive, HBase.

ShineTeck - Hyderabad, India

03/2019-11/2021

Role: Software Engineer

Responsibilities:

- Developed and maintained RESTful APIs using Python and Flask, supporting high-throughput workloads.

- Implemented secure authentication, authorization, and input validation mechanisms for APIs.
- Delivered new API features with a focus on scalability, maintainability, and enterprise integration.
- Optimized API performance through code refactoring, database query tuning, and caching strategies.
- Automated operational workflows using Python scripts to reduce repetitive manual tasks.
- Implemented structured logging and exception handling mechanisms in REST APIs to improve observability, debugging, and production issue resolution.
- Designed and optimized database interactions using efficient SQL queries and connection management to support high-performance API workloads.
- Supported API versioning, backward compatibility, and documentation standards to ensure seamless integration.
- Wrote and maintained unit and integration tests to ensure consistent code quality.
- Utilized API Gateway to manage API traffic, throttling, and monitoring.
- Built dashboards and visualizations to monitor API usage, latency, and performance metrics.
- Configured automated alerts and notifications to improve incident response times.
- Participated in code reviews and collaborated with cross-functional teams to improve API design, security standards, and overall code quality.
- Collaborated with DevOps teams to deploy, monitor, and support APIs in production environments.
- Designed data quality metrics and monitoring dashboards to track data health.
- Supported data governance and standardization initiatives across teams.
- Conducted data mining and trend analysis to support data-driven business decisions.
- Participated in code reviews and supported ETL enhancements and optimizations.
- Automated recurring reports, saving approximately 5 hours per week in manual effort.

Environment: Python, Flask, REST APIs, SQL.

CERTIFICATIONS:

- **Meta Database Engineer**
- **Microsoft Azure Data Engineering Associate**